Business Context

2Market is a global supermarket operating across eight countries. As pressures in retail intensify, the business needed a clearer understanding of who their customers are, what drives purchasing decisions, and which marketing investments yield the greatest return. This project was structured around the Cross-Industry Standard Process for Data Mining (CRISP-DM) model. It shows progress through business understanding, data understanding, data preparation, modelling, evaluation, and deployment. There were three business questions guiding the project: What are the defining demographics of 2Market's customer base? Which advertising channels are most effective at driving conversions? and Which product categories generate the most revenue, and does this vary by customer demographic? To sharpen the scope, key questions were raised at the beginning. These included what precisely constitutes a successful lead conversion given the multiple conversion-tracking fields in the data, whether the six product categories represent 2Market's full product range or a subset and who the intended audience for the final dashboard would be. This would determine which insights to prioritise and the appropriate level of technical depth.

The Analytical Approach

The dataset contained 2,216 customer records across eight countries. The analysis began in Excel during the data preparation phase of CRISP-DM. There were several quality issues that were identified and resolved. The Income field was stored as a formatted text string with dollar signs and commas. As a result, a SUBSTITUTE-VALUE formula was used to produce a clean Income_Clean column, enabling effective analysis. Year_Birth was then converted to an Age column by subtracting from the reference year 2014, consistent with the data collection period, revealing a customer age range of 17 to 120 years. The upper extreme (age 120) represents a clear data entry error and was flagged as an outlier. The Marital_Status field contained seven entries across three invalid categories ("Absurd", "Alone", and "YOLO"). These were consolidated into a cleaned column for analysis to take place. Here, AVERAGEIF and AVERAGEIFS formulas were applied to produce statistics such as average income by band and average age by marital status, informing initial customer profiling.

In SQL, the marketing_data and ad_data tables were joined using a LEFT JOIN on the shared ID column. An Inner Join was considered, however this was rejected as it would have excluded customers with no corresponding advertising data, risking a biased sample. Three queries were then executed: the first aggregated social media conversions by country to identify the most effective platform per market. The second broke this down by marital status to explore whether relationship status influenced channel preference and the third and final query compared total social media ad activity against total customer spend by country to test whether advertising investment correlates with revenue. These cross-table queries produced findings that couldn't be achieved through Excel alone and formed the quantitative basis for the dashboard to be created later on.

Dashboard Development and Design

The dashboard was designed and created in Tableau. This dashboard was designed with a non-technical marketing stakeholder audience in mind. The guiding principle was clarity over complexity. For this reason, each visualisation was chosen to answer a specific business question rather than to demonstrate analytical range. The layout follows a deliberate top-to-bottom narrative structure. The upper section presents a Sales Map, an Age Distribution histogram, and a Sales by Product horizontal bar chart. The histogram was chosen for its ability to reveal the shape of the customer age distribution clearly, while the horizontal bar chart allows quick comparison of spend across product categories for non-specialist audiences. The middle section addresses advertising effectiveness and product spend by age group, using a stacked bar chart and grouped bar chart respectively. Both are chosen to enable comparison across segments without requiring interaction.

The dashboard's primary interactive element is the Sales Map, which functions as a global filter. By clicking a country, all charts simultaneously update. This allows users to explore regional patterns in demographics, product preferences, and ad channel performance effectively. A country dropdown filter also enables more granular control over the Sales by Product view specifically. This supports the adoption principle that dashboards should enable self-service insight rather than function as static reports.

One limitation of the current design is that some countries, particularly in the Middle East, have very small customer samples, which may produce misleading averages. A future iteration could suppress or flag markets below a minimum sample threshold to prevent misinterpretation.

Findings and Insight

In regard to product performance, liquor (Amt Liq) was the dominant category by a considerable margin. It can be seen representing the highest spend across every age group without exception. Non-vegetarian products were consistently second across all demographics. Together, alcohol and meat account for the majority of total product revenue, while vegetables, fish, and chocolates barely register across any age group. The 35–49 age group is the highest-spending segment across every product category, making it 2Market’s most commercially valuable demographic. The 65+ group spends notably little despite potentially having higher disposable income. This might suggest the business may be failing to engage customers of the demographic effectively.

We can then turn to geographic performance. Here, Spain recorded the highest total customer spend at £659,557, while the Middle East recorded the lowest at £3,122. This statistically demonstrates a 211-fold difference. We also see that Spain accounts for nearly half of all total spend across all markets. Australia and Canada show moderate spend and could represent growth opportunities worth further investment.

On advertising effectiveness, Twitter and Bulkmail were the highest-performing channels overall based on the Ad Effectiveness chart. These results were followed by Instagram and Facebook behind them. From the data it is visible that brochure advertising was the least effective source in this category.

Three areas are recommended for follow-up: investigating why the Middle East shows such low spend, this may reflect a data capture gap rather than genuinely low activity, exploring whether the 65+ engagement gap reflects a targeting or product relevance issue and reviewing whether low-performing product categories such as vegetables and fish require promotional intervention or pricing changes.